SUNFALSER
OPTOELECTRONIC SOLUTIONS

SFA2000C

Laser Rangefinder Module

Product Manual

Version 202606V1

Disclaimer

This product manual is intended to provide reference specifications and performance parameters of the product. If certain specifications are not included in this document, please contact SunFlaser Technology (Chengdu) Co., Ltd. directly for further confirmation. All contents of this manual and the specifications of the ranging module are subject to change without prior notice.

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At the time of publication, all descriptions, specifications, designs, and related procedures contained herein are considered valid. SunFlaser Technology (Chengdu) Co., Ltd. reserves the right to modify the above information at any time without prior notice. If you have any questions regarding the use of this manual, please contact us promptly.

Safety Instructions

For the safe use of the ranging module, please carefully follow the safety instructions provided in this manual. Do not attempt to disassemble, modify, or repair any part of the ranging module. Do not tamper with or attempt to alter the performance of the ranging module.

Intended Use

- Take precautions against electrostatic discharge (ESD). Do not touch any electronic components or parts of the ranging module without proper protective measures.
- Keep this manual readily accessible to all personnel operating the ranging module.
- Operate the ranging module only within the specified voltage and power range.
- Operate the ranging module in a clean, dry, and ESD-protected environment.
- Do not touch the optical glass lens with your fingers.
- Avoid exposing the ranging module to environments with drastic temperature fluctuations.

Improper Use

- Using the ranging module with the host system before verifying that all functions are operating properly;
- Operating the ranging module without first reading and understanding this manual and all safety instructions;
- Disassembling, modifying, or altering the ranging module without authorization from SunFlaser Technology (Chengdu) Co., Ltd.;
- Sending operating commands other than those specified in this manual;
- Using or testing the ranging module in non-standard, abnormal, or unsuitable environments;

- Operating the ranging module in a manner inconsistent with the instructions provided in this manual;
- Continuing to operate the ranging module after improper handling or misuse;
- Using a ranging module with obvious faults, defects, or damage;
- Performing test operations within the blind zone of the ranging module.

Improper use may result in the following risks:

- Eye injury
- Incorrect results
- Measurement errors
- Property damage
- Equipment malfunction

Operator Responsibilities

- Be familiar with the basic knowledge required for the correct operation of the ranging module;
- Understand and comply with the operation and maintenance requirements of the ranging module;
- Prevent damage to the ranging module during storage and transportation;
- Keep the ranging module clean and well maintained;
- Ensure that the ranging module is stored and operated in a safe environment;
- Do not expose the ranging module to strong mechanical shock or impact;
- Do not expose the ranging module to environments with drastic temperature fluctuations.

Avoid Measurement Errors

- Pay attention to all factors that may affect the performance of the ranging module;

- After the laser rangefinder module has been subjected to improper handling (such as vibration, impact, or dropping), or before performing critical measurement tasks, it is recommended to verify the performance specifications of the ranging module.

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01 Product Overview

The SFA2000C is a highly integrated miniature laser ranging module based on Time-of-Flight (TOF) technology. It features miniaturization, lightweight design, low power consumption, and high stability and reliability. It is suitable for UAVs, robotics, micro gimbal pods, handheld devices, and various intelligent electro-optical platforms.

The product adopts a 905 nm laser solution and communicates via a TTL serial interface, supporting multiple ranging modes for easy and rapid integration into flight control systems, embedded platforms, and a wide range of smart devices. Its modular structural design ensures stable ranging performance while further optimizing size and weight, making it especially suitable for space- and payload-constrained applications.

The SFA2000C combines long-range measurement capability with fast response performance, meeting target ranging requirements in complex outdoor environments and making it suitable for target measurement, auxiliary perception, intelligent control, and lightweight electro-optical system integration applications.

02 Key Features

> Micro Lightweight Design

Adopting a highly integrated modular structural design, the module features a compact size and lightweight construction, making it well-suited for integration into lightweight platforms such as UAVs, robots, micro gimbal systems, and portable devices.

> Long-Range Stable Distance Measurement

Based on Time-of-Flight (TOF) ranging technology, the module enables stable and reliable distance measurement, balancing long-range capability with performance in complex environments.

> High-Precision Measurement

The module delivers excellent ranging accuracy and repeatability, providing stable output of target distance data to meet the requirements of target measurement and auxiliary perception applications.

> High-Speed Ranging Support

Supports multiple ranging frequency modes, enabling real-time measurement and dynamic target tracking, suitable for high-response scenarios.

> Flexible Measurement Modes

Supports both single-shot and continuous ranging modes, allowing flexible switching according to different application requirements, suitable for both static and dynamic targets.

> Low Power Consumption Operation

Designed with low power consumption architecture, suitable for battery-powered devices and long-duration operation scenarios, helping to extend overall system endurance.

> Standard Serial Communication

Adopts UART-TTL communication interface with a simple protocol, enabling easy integration with flight controllers, microcontrollers, embedded platforms, and host computer systems.

> Strong Environmental Adaptability

Designed with robust environmental adaptability, ensuring stable operation in complex outdoor environments and industrial application scenarios.

> Modular and Easy Development

Provides complete communication protocol and development support, facilitating system integration and secondary development, and significantly reducing integration complexity.

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Key Specifications

No.	Item	Performance Specification
1	Operating Wavelength	905 nm
2	Transmit Aperture	Ø10 mm
3	Receive Aperture	Ø15 mm
4	Ranging Capability	≥2000 m (large target; visibility ≥10 km; humidity ≤60%; target reflectivity: 0.3)
		≥1000 m (human target or equivalent; visibility ≥10 km; humidity ≤60%; target reflectivity: 0.3)
5	Blind Zone	≤3 m
6	Ranging Accuracy	≤±0.5 m (≤80 m), ±1 m (≤1000 m), ≤±(0.2 + 0.0015 × D) (>1000 m)
7	Divergence Angle	≤6 mrad
8	Accuracy Rate	≥99%
9	False Alarm Rate	≤2%
10	Ranging Frequency	Single-shot, 1 Hz (≥2000 m), 5 Hz (≥1000 m), 10 Hz (function supported)
11	Baud Rate	115200 bps
12	Weight	≤12 g
13	Dimensions	≤26 mm×25 mm×13.5 mm
14	Electrical Interface	Connector plug: 0.8WTB-6Y-2
		Connector receptacle: 0.8WTB-6AWB-01
15	Supply Voltage	3.3 V to 5 V
16	Standby Power Consumption	≤0.8 W @ 1 Hz
17	Average Power	≤1.6 W @ 1 Hz

	Consumption	
18	Operating Temperature	-20°C to +65°
19	Storage Temperature	-45°C to +70°C

Note 1: Items marked with * are design values or theoretical calculated values for reference only.

Note 2: The nominal ranging distance and accuracy are achieved under conditions of visibility not less than 10 km and relative humidity not greater than 60%. The ranging capability and accuracy may vary depending on environmental conditions and target characteristics; actual performance shall prevail based on real measurements.

When the module is used on a stabilized platform, ensure that the platform can stably track the target; otherwise, it may affect the ranging distance or accuracy of the module.

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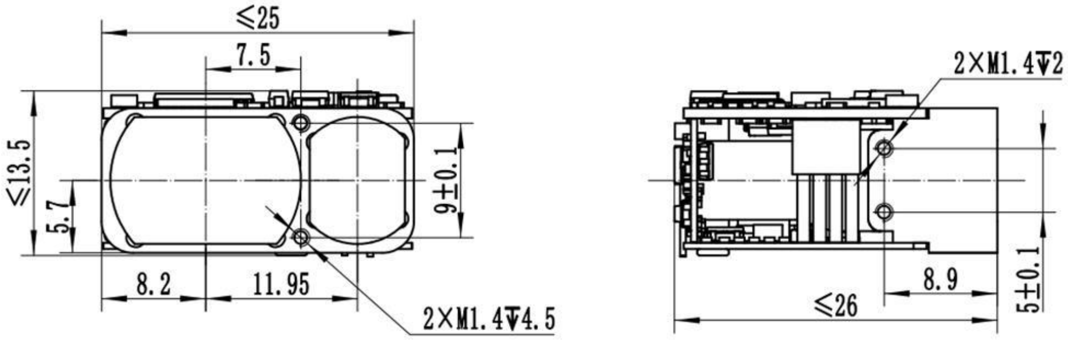
Interface and Connection

Mechanical & Optical Interface:

The mechanical and optical interface dimensions of the laser ranging module are shown in the figure below, unit: mm.



Technical Requirements:

1.Unit: mm;

2.Unless otherwise specified, tolerances follow GB/T 1804-2000, grade m.

Electrical Interface:

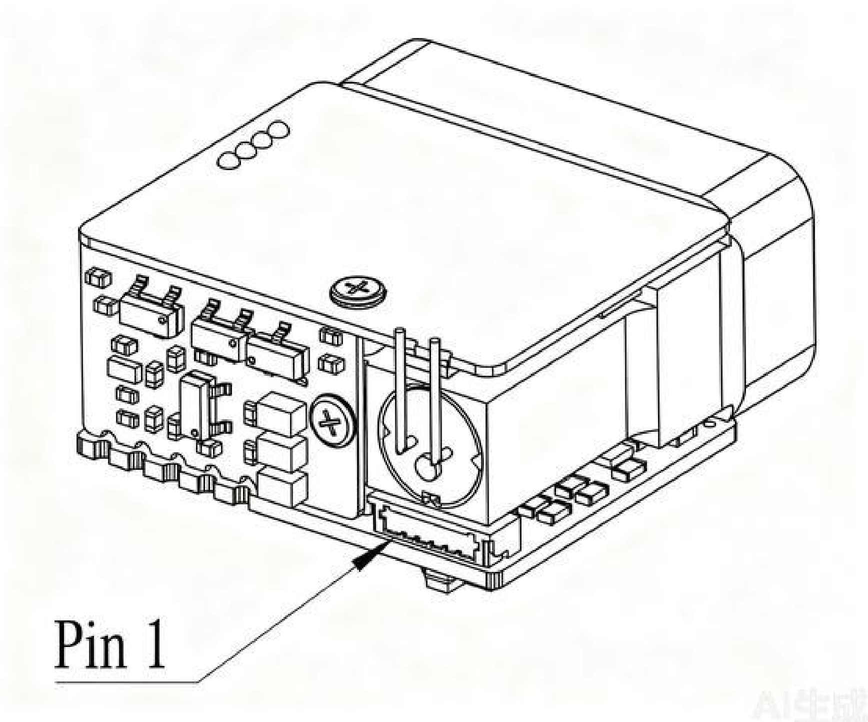
The electrical interface requirements are as follows:

- Supply Voltage: 3.3 V – 5 V
- Standby Power Consumption: ≤0.8 W @ 1 Hz
- Average Power Consumption: ≤1.6 W @ 1 Hz

The host system connects to the laser ranging module via a 0.8WTB-6Y-2 connector, mating with the module-side 0.8WTB-6AWB-01 connector (Yueqing Huabao) for interconnection testing.

The power supply and communication interface pin definitions on the module side are shown in the table below.

Pin No.	Label	Electrical Definition	Signal Direction
1	Power-EN		
2	TTL_RXD	Signal input port	Host to module
3	TTL_TXD	Signal output port	Module to host
4	NC		
5	VCC / Power positive		
6	GND		



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Communication Protocol and Commands

For details of the following contents, please refer to Appendix 1:

- Communication Interface
- Single Measurement Command
- Continuous Measurement Command
- Stop Measurement Command

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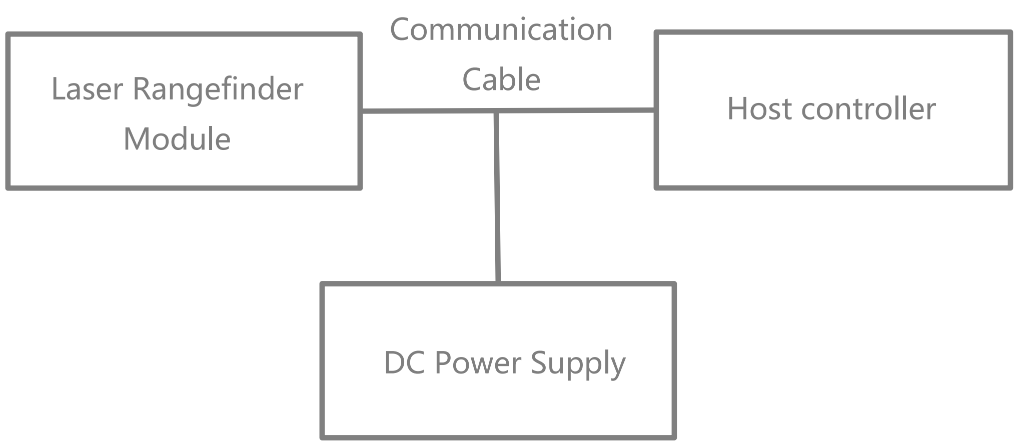
Operating Instructions

Power-On Procedure

- Before powering on, connect the laser rangefinder module, communication cable, DC power supply, and host computer according to the electrical interface requirements and as shown in the figure below.



- The module powers on automatically after the power supply is connected.

Power-Off Procedure

- Before powering off, ensure that all operating processes and tasks have been completed and that the program has been exited properly.
- After confirmation, disconnect the power supply to power off the module normally.

Ranging Mode Operation

- Send the "Single Ranging" command to the laser rangefinder module The module performs a single ranging operation and returns the ranging status and distance value.
- Send the "Continuous Ranging" command to the laser rangefinder module. The module performs continuous ranging and returns the ranging status and distance value continuously.
- Send the "Stop Ranging" command to terminate the ranging operation.

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Precautions for Use

- The laser ranging module should not be viewed directly by the human eye. If observation is necessary, it must be performed with appropriate professional protective equipment. Direct exposure to the laser beam is strictly prohibited to avoid potential eye injury.
- Avoid ranging on targets within the blind zone, especially highly reflective surfaces such as glass or polished metal. Avoid operating multiple modules face-to-face at close range. Do not allow high-energy laser sources to directly irradiate the receiving optics. During assembly and debugging, the receiving lens must remain covered at all times; otherwise, excessively strong echo signals may cause permanent damage to the detector.
- ESD protection must be observed: the electronic components of the laser ranging module are electrostatic discharge sensitive devices. Do not touch any electronic components without proper protection.
- Do not touch the optical lenses with fingers or hard objects. Avoid cleaning the lenses unless necessary. Do not use cleaning agents unless specified, as this may cause irreversible damage to the optical coating and affect performance.
- Do not use the product under direct sunlight. Do not store it in highly polluted environments or outside the specified storage temperature range. Avoid strong mechanical impacts such as vibration, shock, or dropping. Do not expose the module to rapid temperature changes, as this may lead to unpredictable performance issues.
- The module adopts a certain level of sealing design; however, it is recommended to operate it in environments with humidity below 80% and maintain a clean working environment to extend service life.

- Do not attempt to disassemble, modify, or repair any part of the module. Do not attempt to tamper with or adjust its performance.
- Operate the module strictly according to the instructions in this specification. Do not send any control commands that are not defined in this document.

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Optical Window Requirements

Material Recommendations

The optical window material is recommended to be Chengdu Guangming Optical Glass H-K9L or fused silica. H-K9L and fused silica are commonly used colorless optical glasses suitable for laser wavelengths from 300 nm to 2100 nm. They offer high cost-effectiveness and excellent physical properties.

Manufacturing Recommendations

The wedge angle tolerance of the optical window should be as small as possible. It is recommended that the wedge angle tolerance be ≤ 3′ (tolerance grade ≤ 7). The optical surface of the window should be as smooth as possible, with a recommended arithmetic average roughness (Ra) of 0.012 μm.

Coating Recommendations:

For 905 nm laser rangefinder modules, it is recommended to apply an anti-reflection (AR) coating optimized for the 855 nm–955 nm wavelength range, with a transmission rate of ≥99.5%.

Depending on the specific application environment, additional protective coatings such as hydrophobic coatings or hard coatings may be applied to the outer surface of the optical window. Other performance requirements should comply with GJB 2485-95, with a transmission rate of ≥98%.

> Optical Window Shape and Usage Recommendations:

The effective aperture of the optical window varies depending on different products. Its overall dimensions should satisfy the following requirements: the difference between the optical window effective aperture and the outer diameter should be ≥ 2 mm; and the difference between the rangefinder antenna outer diameter and the projected size of the optical window effective aperture should be ≥ 1.5 mm, as shown in the schematic diagram below.

Since the optical window has a certain level of laser absorption, it is recommended that the thickness of the optical window be controlled within 2–4 mm according to the overall size.

Due to the high transmittance of the optical window, it is recommended that the emission optical axis be aligned parallel to the normal of the optical window surface. The positioning relationship between the optical window and the dual optical barrels is shown in the schematic diagram below.

Meanwhile, the air gap between the optical window and the rangefinder should be less than 0.5 mm. The figure below illustrates two installation configurations of the optical window.

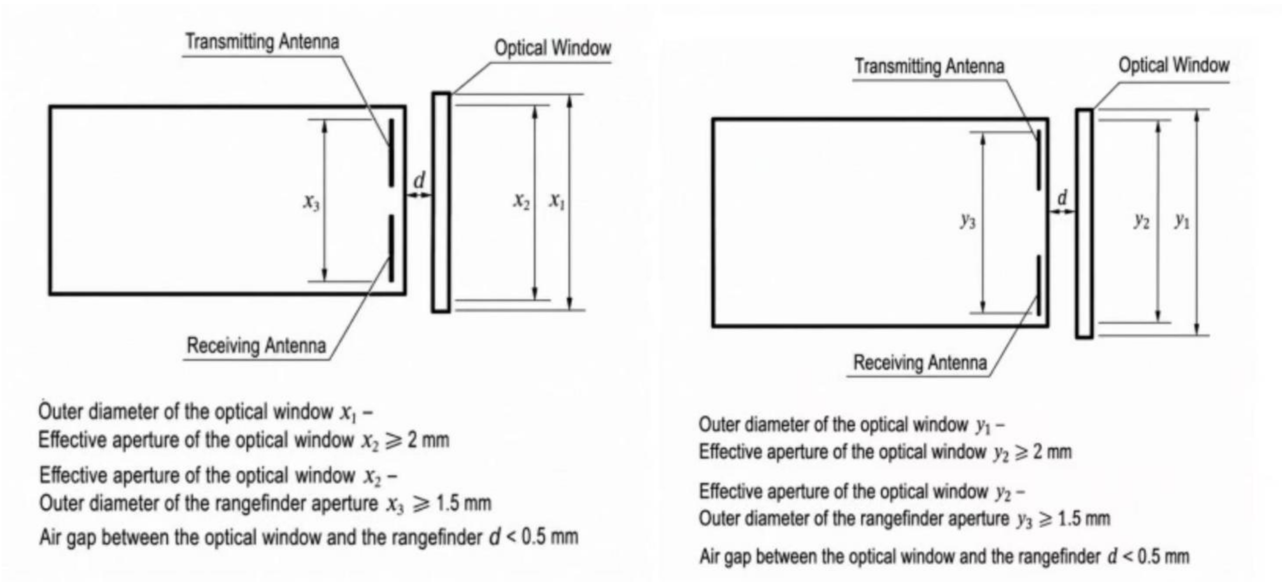

Figure 1: Optical Window Dimensions and Placement Diagram

09 Safety Precautions

- To ensure the safety of both operators and test subjects, the following safety measures have been incorporated into the design of the SFA2000C laser rangefinder module;
- Safety design and analysis are conducted in accordance with GJB 900A-2012, General Requirements for Equipment Safety;
- Non-flammable materials are used, and all mechanical and electrical interfaces are securely connected;
- Appropriate design measures are implemented to prevent moisture accumulation that could result in short circuits;
- The module operates below the safety voltage limit for the human body.

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Maintenance Instructions

Packing List

The standard items supplied with the product are listed in the table below. Optional accessories may not be included in this list and shall be subject to the actual product delivered.

No.	Item	Quantity	Remarks
1	Laser Rangefinder Module	1 unit	-
2	Certificate of Conformity	1 copy	-
3	Packaging Box	1 pc	-
4	Connector plug	1 pc	-

Module Cleaning

Cleaning of Optical Components

- Dust particles should be removed using an air blower;
- Fingerprints should first be cleaned with absorbent cotton lightly moistened with a small amount of alcohol-ether mixture, and then wiped with a clean lens cloth.

Cleaning of Mechanical and Electronic Components

- With the power disconnected, gently wipe mechanical and electronic components with alcohol and allow them to dry naturally before use;
- Keep the ranging module, connectors, and cables away from moisture and contaminants whenever possible;

- Ensure that the equipment is completely dry before packaging.

Inspection and Maintenance

Routine Maintenance

When the product is used for the first time, or after replacement of related modules or components, both visual inspection and power-on inspection shall be performed. For products under normal operation, only a power-on inspection is required before use.

- Visual Inspection Procedure
  - Check whether the product appearance is normal;
  - Check whether the cable connections are correct and securely connected.
- Power-On Inspection Procedure
  - Perform the power-on procedure and verify that the standby power is correct;
  - Start the self-test function of the product;
  - After the inspection is completed, perform the power-off procedure.

Periodic Maintenance

Under normal operating conditions, the laser rangefinder module does not require routine maintenance. If the module has been stored in a dust-free environment for more than one year, maintenance should be performed, including general inspection and power-on inspection.

- General Inspection (Performed with Power Off)
  - All markings and serial numbers on the product and test cable connectors shall be correct and clearly legible;
  - All screws on the panel shall be securely fastened;
  - The optical glass of the product shall be free from visible defects that may affect normal operation, such as spots, pitting, water stains, fungus, fingerprints, dust particles, or cracks.
- Comprehensive Power-On Inspection and Maintenance

- Connect the product power supply in sequence;
- Perform the power-on procedure and verify that the standby power is correct;
- Start the product self-test and complete the self-test procedure;
- Perform the power-off procedure after completion of the inspection.

Troubleshooting

The following list only includes possible fault phenomena not caused by the product itself. If any other technical problems occur during operation, please contact our company for technical support.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
No communication or abnormal communication of the rangefinder module	Power supply issue	Check whether the power supply has hardware or software wake-up enabled. Verify that the voltage setting is correct and ensure the protection current is properly configured. If the voltage is correct, try replacing the power supply and test again.	If the module operates normally after changing the voltage or power supply, the issue can be determined to be caused by an incorrect power supply.	Set the correct voltage and current, or replace the power supply.
	Connection cable issue	Check whether the wiring is secure, or replace the communication cable for testing (TTL communication communication cables should not be excessively long).	If communication is restored after rewiring or replacing the communication cable, the issue can be determined to be caused by a connection cable problem.	Rewire the connections or replace the communication cable.
		Check whether the wiring matches the interface pin definition.	If the module operates normally after confirming that the wiring matches the pin definition, the issue can be determined to be caused by incorrect wiring.	Reconnect the wiring according to the correct pin definition.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
	Incorrect baud rate setting	1. Check whether the rangefinder status and baud rate settings are consistent with the communication protocol. 2. If the rangefinder supports baud rate configuration, change to the required baud rate and try communication again.	If communication operates normally after aligning the baud rate with the communication protocol, the issue can be determined to be caused by an incorrect baud rate setting.	Set the host computer baud rate to match the communication protocol.
	Incorrect communication command	Check the rangefinder status and verify whether the transmitted command is consistent with the communication protocol, and confirm that the command is in hexadecimal format.	If communication operates normally after confirming that the transmitted command matches the communication protocol, the issue can be determined to be caused by an incorrect communication command.	Send the correct command to the rangefinder module.
	RS422 driver not installed on the host computer.	Install the RS422 driver on the host computer.	If communication with the rangefinder module works normally after installation, the issue can be determined to be caused by the missing driver installation.	Install the driver.
	Main control program issue	If communication fails during integration with the main controller, first check whether standalone communication is functioning normally.	If standalone communication works normally, the issue can be considered likely to be caused by a problem in the main control program.	It is recommended to check whether there are issues in the system-level program (e.g., whether the parity/check bits are correct, and whether the host computer data parsing and display are implemented correctly).
	Other causes (e.g., circuit failure, etc.)	Return to factory for inspection and troubleshooting.	Return to the factory for inspection and diagnosis.	Return to factory for repair.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
The rangefinder returns invalid values.	Operational error / improper operation.	Check whether the receiving or transmitting lens is blocked, and ensure that there are no obstructions in the measurement environment (e.g., window glass, wall corners, etc.).	If the rangefinder returns normal readings after removing the obstruction, the issue can be considered likely caused by optical blockage.	Remove the obstruction.
		Whether the ranging axis is aligned with the optical sighting axis.	If the rangefinder returns normal measurements after realignment, the issue can be determined to be caused by misalignment between the ranging axis and the optical sighting axis.	Realign the ranging and optical axes and perform measurement again.
	Laser transmitter failure.	Check whether the current reading is normal. Observe whether the current fluctuates during ranging. Send a self-test command to check the returned data. Alternatively, use an infrared test card or a CCD camera to verify whether a laser spot is present.	If the emission self-test is abnormal and no laser spot is observed, the issue can be preliminarily determined to be caused by a laser transmitter failure.	Return to factory for repair.
	Other causes (e.g., circuit failure, detector, etc.)	Return to factory for inspection and troubleshooting.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
False ranging results.	Electrical noise interference.	Block or point the rangefinder receiving lens toward open space (or away from the sun) and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by optical noise interference. Further determine whether it is due to external ambient light or the system optical window (e.g., mismatch of wavelength band with the rangefinder, excessive tilt angle of the optical window, small aperture causing light blockage, or a contaminated optical window).		It is recommended to address power supply noise issues by adding a larger common-mode filter in series, replacing the grounding capacitor with a 10 nF capacitor, and increasing the detection threshold (note that this may reduce system performance).

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
	Communication noise interference.	Power the rangefinder module independently within the system and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by electrical or communication noise interference. Power the rangefinder module independently within the system and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by power supply noise interference.		It is recommended to address communication noise interference by adding a ferrite core (shielding magnetic ring) to the cable or increasing the threshold level (note that this may reduce system performance).
	Optical noise interference.	Operate the rangefinder module with communication only within the system and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by communication noise interference. If false alarms still occur after the above steps, test the rangefinder module as a standalone unit (removed from the system and powered/communicated independently). If false alarms are still present, the issue can be determined to be caused by the rangefinder module itself or the power adapter.		It is recommended to address optical noise issues (optical window condition and test angle) and increase the detection threshold (note that this may reduce system performance).
	Other causes (e.g., circuit failure, etc.).	Return to factory for inspection and troubleshooting.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
Insufficient performance capability of the rangefinder module.	Environmental conditions (weather-related factors).	Check whether the weather conditions include rain, snow, fog, or dust/sand, whether atmospheric visibility is too low, whether humidity is too high, or whether there are localized atmospheric turbulence effects, as well as any discrepancies between the environmental conditions and the test site.	If the ranging performance meets the technical specifications when the visibility is above the required threshold, the issue can be determined to be caused by environmental weather conditions.	Test the rangefinder performance under appropriate weather conditions.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
	Optical window issue.	Determine, by checking the system configuration, whether the issue is caused by low optical window transmittance, incorrect wavelength band, contamination, small aperture causing light blocking, or excessive tilt angle. Remove the optical window and test the rangefinder module independently to observe whether its ranging performance is degraded. Also inspect whether the lens of the rangefinder is excessively contaminated.	If the ranging performance meets the specification requirements after removing the optical window, the issue can be determined to be caused by low transmittance of the host optical window. If the ranging performance meets the specification requirements after cleaning the rangefinder lens, the issue can be determined to be caused by low transmittance of the host optical window.	Replace the optical window according to the recommended selection and installation guidelines, or switch to a higher-performance rangefinder model. Clean the lens.
	Target size is too small or target reflectivity is too low.	Replace with another target. It is recommended to use targets such as buildings that are perpendicular to the optical axis, and avoid hollow or perforated targets as much as possible.	If the ranging performance meets the specification requirements after replacing with a higher-reflectivity or larger target, the issue can be determined to be caused by inappropriate target selection.	Replace the target, or switch to a higher-performance rangefinder model.
	Operating temperature exceeds the specified range of the rangefinder module.	When ranging performance degrades, send a self-test command to check the current operating temperature of the rangefinder module. After the temperature returns to normal, retest and observe whether the performance issue persists.	If the operating temperature exceeds the specified range when performance degrades, and the performance issue disappears after the temperature returns to normal, the issue can be determined to be caused by the operating environment exceeding the allowable temperature range of the rangefinder module.	Operate the device within the specified operating temperature range.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
	Optical axis misalignment.	Check whether high/low temperature tests, thermal shock tests, shock tests, or vibration tests were performed prior to the degradation of ranging performance, or whether any drop or impact event occurred.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
	Other causes (e.g., circuit failure, low receiver sensitivity, etc.)	Return to factory for inspection and troubleshooting.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
Accuracy out of tolerance.	Large variation in target reflectivity.	Confirm whether the tested target is a high-reflectivity target, and determine whether the issue is large data fluctuation or accuracy out of tolerance.	If the tested target is highly reflective and the ranging accuracy returns to normal after replacing it with a suitable target, the issue can be determined to be caused by excessive target reflectivity variation.	Replace the target. The use of highly reflective test targets is not recommended.
	Test operation issue.	Check whether the target has been properly aimed at.	If the ranging accuracy returns to normal after correctly aiming at the target, the issue can be determined to be caused by improper test operation.	Re-aim at the target.
	Instrument issue.	Verify the accuracy of the calibration instrument (possible calibration error).	If the ranging accuracy returns to normal after replacing with an accurately calibrated instrument, the issue can be determined to be caused by the test instrument.	Replace the calibration instrument.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
	Low optical window transmittance.	Check whether the ranging value is abnormally high.	If the ranging accuracy returns to normal after removing the optical window, the issue can be determined to be caused by the optical window.	Replace the optical window according to the recommended optical window selection and installation guidelines.

Packaging and Transportation Requirements

Packaging

After unpacking, if the product needs to be placed back into storage, it shall be repackaged using the original packaging whenever possible.

If the product needs to be returned to the manufacturer, the original packaging is recommended. If alternative packaging is used, it shall not cause any degradation in product performance or physical damage.

Transportation

Repackaged products may be transported by automobile, railway, aircraft, or ship. During transportation, the package shall be securely fixed to the transport vehicle to avoid impact, improper handling, and exposure to rain or snow.

11 Technical Support and Service

If a product malfunction occurs and factory return is required for fault analysis, troubleshooting, or repair, SunFlaser Technology (Chengdu) Co., Ltd. provides a one-year warranty and lifetime technical support from the date of product delivery.

During the warranty period, product failures caused by manufacturing or quality issues will be repaired or replaced free of charge by SunFlaser. For product damage caused by improper operation or other user-related factors, repair and replacement costs will be charged according to the actual situation.

This product is a precision optical instrument. Please handle and operate it carefully during use. If you have any additional questions regarding operation or maintenance, please contact our after-sales support personnel at any time.

12 Contact Information

Website: www.sunflaser.com

Tel: +86 186 2827 7862 (Sales Manager: Mr. Cao)

Address: No. 58, Zhenxing West 1st Road, Building 1, Room 201, Jinniu District, Chengdu, Sichuan, China

Appendix 1:

Communication Interface

- Baud rate: 115200 bps;
- Single-byte transmission format: 1 start bit, 8 data bits, no parity, and 1 stop bit; 8-bit data is transmitted with the low byte first, then the high byte.

1 Communication Protocol

1.1 Command 1

1.1.1 Single Ranging Command

Note: Transmit checksum = Byte 3 + Byte 4 + Byte 5 + Byte 6 + Byte 7;

Receive checksum = Byte 1 + Byte 2 + Byte 3 + Byte 4 + Byte 5 + Byte 6 + Byte 7.

Command sent to the ranging module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x88	0xFF	0xFF	0xFF	0xFF	0x84

Response from the ranging module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x88	status	0xFF	DATA_H	DATA_L	checksum

Status = 0 indicates single measurement failure; DATA_H = 0xFF, DATA_L = 0xFF;Status = 1 indicates single measurement success; DATA_H = high byte of measurement result; DATA_L = low byte of measurement result.

1. 1. 2 Continuous Ranging Command

Note: Transmit checksum = Byte 3 + Byte 4 + Byte 5 + Byte 6 + Byte 7;

Receive checksum = Byte 1 + Byte 2 + Byte 3 + Byte 4 + Byte 5 + Byte 6 + Byte 7.

Command sent to the ranging module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	Freq	0xFF	0xFF	0xFF	0xFF	checksum

Response from the ranging module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	Freq	status	0xFF	DATA_H	DATA_L	checksum

Status = 0 indicates continuous measurement failure; DATA_H = 0xFF, DATA_L = 0xFF;Status = 1 indicates continuous measurement success; DATA_H = high byte of measurement result; DATA_L = low byte of measurement result.

Freq = 0x89 for 1 Hz ranging; 0xB9 for 5 Hz ranging; 0xC9 for 10 Hz ranging; Freq = 0xF9 for calibration mode (returns calibration status once upon receiving calibration command).

1. 1. 3 Stop Measurement

Command sent to the ranging module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x8E	0xFF	0xFF	0xFF	0xFF	0x8A

Response from the ranging module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x8E	status	0xFF	0xFF	0xFF	checksum

Status = 0 indicates stop continuous measurement failure; status = 1 indicates stop continuous measurement success.

Note: All returned data is in hexadecimal format, and all distance values are multiplied by 10 before output.

Example: dist = 2000.3 m → output = 20003 → hex = 0x4E23, where Data1 = 0x4E and Data2 = 0x23.

1. 2 Command 2

1. 2. 1 Information

The control command format is shown in Table 1.

Table 1 Control Commands Received by the Module

Byte	Description	Byte Value	Remarks
1	Frame header	0x55	
		0x02	Single
		0x03	1 Hz
		0x04	5 Hz
		0x07	10 Hz
		0x05	self-test
		0x0A	calibration
		0x00	stop
3	Frame tail	0xAA	

1. 2. 2 Module Response Data

Module response data is shown in Table 2, and self-test data is shown in Table 3.

Table 2 Module Response Data

Byte	Description	Value
1	Frame header	0xAA
2	Integer high byte of distance	
3	Integer low byte of distance	
4	Decimal part of distance	0x00 - 0x63
5	Reserved	Reserved
6	Frame tail	0x55

Table 3 Self-test Data

Byte	Description	Value
1	Frame header	0xAA
2	Temperature information	
3	Reserved	Reserved
4	Reserved	Reserved
5	Reserved	Reserved
6	Frame tail	0x55

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